Constructions With Relations

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This chapter contains some material about constructions with relations. Notably, we discuss and explore:

1. The existence or non-existence of Kan extensions and Kan lifts in the 2-category **Rel** (Section 9.2).
2. The various kinds of constructions involving relations, such as graphs, domains, ranges, unions, intersections, products, converse relations, composition of relations, and collages (Section 9.3).

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9.1 Co/Limits in the Category of Relations

This section is currently just a stub, and will be properly developed later on.

9.2 Kan Extensions and Kan Lifts in the 2-Category of Relations

9.2.1 Left Kan Extensions in Rel

PROPOSITION 9.2.1.1.1 ▶ LEFT KAN EXTENSIONS IN Rel

Let $R\colon A \nrightarrow B$ be a relation.

1. *Non-Existence of All Left Kan Extensions in* **Rel**. Not all relations in **Rel** admit left Kan extensions.

2. *Characterisation of Relations Admitting Left Kan Extensions Along Them.* The following conditions are equivalent:

   (a) The left Kan extension

   $$\mathrm{Lan}_R\colon \mathbf{Rel}(A, X) \to \mathbf{Rel}(B, X)$$

   along R exists.

   (b) The relation R admits a left adjoint in **Rel**.

   (c) The relation R is of the form f^{-1} (as in **??**) for some function f.

Proof 9.2.1.1.2 ▶ Proof of Proposition 9.2.1.1.1

Item 1: Non-Existence of All Left Kan Extensions in Rel

Omitted, but will eventually follow Fosco Loregian's comment on [MO 460656].

Item 2: Characterisation of Relations Admitting Left Kan Extensions A

Omitted, but will eventually follow Tim Campion's answer to to [MO 460656].

Question 9.2.1.1.3 ▶ Existence of Specific Left Kan Extensions of Relations

Given relations $S\colon A \nrightarrow X$ and $R\colon A \nrightarrow B$, is there a characterisation of when the left Kan extension

$$\mathrm{Lan}_S(R)\colon B \nrightarrow X$$

exists in terms of properties of R and S?
This question also appears as [MO 461592].

Question 9.2.1.1.4 ▶ Explicit Description of Left Kan Extensions Along Functions

As shown in Item 2 of Proposition 9.2.1.1.1, the left Kan extension

$$\mathrm{Lan}_R\colon \mathbf{Rel}(A, X) \longrightarrow \mathbf{Rel}(B, X)$$

along a relation of the form $R = f^{-1}$ exists. Is there a explicit description of it, similarly to the explicit description of right Kan extensions given in Proposition 9.2.3.1.1?
This question also appears as [MO 461592].

9.2.2 Left Kan Lifts in Rel

Proposition 9.2.2.1.1 ▶ Left Kan Lifts in Rel

Let $R\colon A \nrightarrow B$ be a relation.

1. *Non-Existence of All Left Kan Lifts in* **Rel**. Not all relations in **Rel** admit left Kan lifts.

2. *Characterisation of Relations Admitting Left Kan Lifts Along Them.* The following conditions are equivalent:

   (a) The left Kan lift

   $$\mathrm{Lift}_R\colon \mathbf{Rel}(X, B) \to \mathbf{Rel}(X, A)$$

   along R exists.

   (b) The relation R admits a right adjoint in **Rel**.

   (c) The relation R is of the form $\mathrm{Gr}(f)$ (as in **??**) for some function f.

Proof 9.2.2.1.2 ▶ Proof of Proposition 9.2.2.1.1

Item 1: Non-Existence of All Left Kan Lifts in Rel

Omitted, but will eventually follow (the dual of) Fosco Loregian's comment on [MO 460656].

Item 2: Characterisation of Relations Admitting Left Kan Lifts Along Th

Omitted, but will eventually follow Tim Campion's answer to to [MO 460656].

Question 9.2.2.1.3 ▶ Existence of Specific Left Kan Lifts of Relations

Given relations $S\colon A \nrightarrow X$ and $R\colon A \nrightarrow B$, is there a characterisation of when the left Kan lift

$$\mathrm{Lift}_S(R)\colon X \nrightarrow A$$

exists in terms of properties of R and S?
This question also appears as [MO 461592].

As shown in Item 2 of Proposition 9.2.2.1.1, the left Kan lift

$$\mathrm{Lift}_R \colon \mathbf{Rel}(X, B) \longrightarrow \mathbf{Rel}(X, A)$$

along a relation of the form $R = \mathrm{Gr}(f)$ exists. Is there a explicit description of it, similarly to the explicit description of right Kan lifts given in Proposition 9.2.4.1.1?
This question also appears as [MO 461592].

9.2.3 Right Kan Extensions in Rel

Let $R \colon A \nrightarrow B$ be a relation.

Proposition 9.2.3.1.1 ▶ Existence of Right Kan Extensions in Rel

The right Kan extension

$$\mathrm{Ran}_R \colon \mathrm{Rel}(A, X) \longrightarrow \mathrm{Rel}(B, X)$$

along R in **Rel** exists and is given by

$$\mathrm{Ran}_R(S) \overset{\text{def}}{=} \int_{a \in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, S_a^{-1}\right)$$

for each $S \in \mathrm{Rel}(A, X)$, so that the following conditions are equivalent:

1. We have $b \sim_{\mathrm{Ran}_R(S)} x$.
2. For each $a \in A$, if $a \sim_R b$, then $a \sim_S x$.

Proof 9.2.3.1.2 ▶ Proof of Proposition 9.2.3.1.1

We have

$$\mathrm{Hom}_{\mathbf{Rel}(A,X)}(S \diamond R, T) \cong \int_{a \in A} \int_{x \in X} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left((S \diamond R)_a^x, T_a^x\right)$$

$$
\begin{aligned}
&\cong \int_{a\in A}\int_{x\in X}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(\left(\int^{b\in B} S_b^x \times R_a^b\right), T_a^x\right)\\
&\cong \int_{a\in A}\int_{x\in X}\int_{b\in B}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x \times R_a^b, T_a^x\right)\\
&\cong \int_{a\in A}\int_{x\in X}\int_{b\in B}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, T_a^x\right)\right)\\
&\cong \int_{b\in B}\int_{x\in X}\int_{a\in A}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, T_a^x\right)\right)\\
&\cong \int_{b\in B}\int_{x\in X}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x, \int_{a\in A}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, T_a^x\right)\right)\\
&\cong \mathrm{Hom}_{\mathbf{Rel}(B,X)}\left(S, \int_{a\in A}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, T_a^{-1}\right)\right)
\end{aligned}
$$

naturally in each $S \in \mathbf{Rel}(B, X)$ and each $T \in \mathbf{Rel}(A, X)$, showing that

$$\int_{a\in A}\mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, T_a^{-1}\right)$$

is right adjoint to the precomposition functor $- \diamond R$, being thus the right Kan extension along R. Here we have used the following results, respectively (i.e. for each $\cong$ sign):

1. Relations, Item 1 of Proposition 8.1.1.1.10.
2. **??**.
3. Ends and Coends, **??** of **??**.
4. Sets, Proposition 3.2.2.1.5.
5. Ends and Coends, **??** of **??**.
6. Ends and Coends, **??** of **??**.
7. Relations, Item 1 of Proposition 8.1.1.1.10.

This finishes the proof. ∎

9.2.4 Right Kan Lifts in Rel

Let $R\colon A \nrightarrow B$ be a relation.

Proposition 9.2.4.1.1 ▶ Existence of Right Kan Lifts in Rel

The right Kan lift

$$\mathrm{Rift}_R\colon \mathrm{Rel}(X,B) \longrightarrow \mathrm{Rel}(X,A)$$

along R in **Rel** exists and is given by

$$\mathrm{Rift}_R(S) \overset{\text{def}}{=} \int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_{-_1}, S^b_{-_2}\right)$$

for each $S \in \mathrm{Rel}(X,B)$, so that the following conditions are equivalent:

1. We have $x \sim_{\mathrm{Rift}_R(S)} a$.
2. For each $b \in B$, if $a \sim_R b$, then $x \sim_S b$.

Proof 9.2.4.1.2 ▶ Proof of Proposition 9.2.4.1.1

We have

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{Rel}(X,B)}(R \diamond S, T) &\cong \int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left((R\diamond S)^b_x, T^b_x\right)\\
&\cong \int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(\left(\int^{a\in A} R^b_a \times S^a_x\right), T^b_x\right)\\
&\cong \int_{x\in X}\int_{b\in B}\int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a \times S^a_x, T^b_x\right)\\
&\cong \int_{x\in X}\int_{b\in B}\int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S^a_x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a, T^b_x\right)\right)\\
&\cong \int_{x\in X}\int_{a\in A}\int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S^a_x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a, T^b_x\right)\right)\\
&\cong \int_{x\in X}\int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S^a_x, \int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a, T^b_x\right)\right)
\end{aligned}$$

$$\cong \mathrm{Hom}_{\mathbf{Rel}(X,A)}\left(S, \int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_{-_1}, T^b_{-_2}\right)\right)$$

naturally in each $S \in \mathbf{Rel}(X, A)$ and each $T \in \mathbf{Rel}(X, B)$, showing that

$$\int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_{-_1}, S^b_{-_2}\right)$$

is right adjoint to the postcomposition functor $R \diamond -$, being thus the right Kan lift along R. Here we have used the following results, respectively (i.e. for each $\cong$ sign):

1. Relations, Item 1 of Proposition 8.1.1.1.10.
2. **??**.
3. Ends and Coends, **??** of **??**.
4. Sets, Proposition 3.2.2.1.5.
5. Ends and Coends, **??** of **??**.
6. Ends and Coends, **??** of **??**.
7. Relations, Item 1 of Proposition 8.1.1.1.10.

This finishes the proof. ∎

9.3 More Constructions With Relations

9.3.1 The Domain and Range of a Relation

Let A and B be sets.

Definition 9.3.1.1.1 ► The Domain and Range of a Relation

Let $R \subset A \times B$ be a relation.[1,2]

1. The **domain of** R is the subset $\mathrm{dom}(R)$ of A defined by

$$\mathrm{dom}(R) \overset{\text{def}}{=} \left\{ a \in A \;\middle|\; \begin{array}{l} \text{there exists some } b \in B \\ \text{such that } a \sim_R b \end{array} \right\}.$$

2. The **range of** R is the subset $\mathrm{range}(R)$ of B defined by

$$\mathrm{range}(R) \overset{\text{def}}{=} \left\{ b \in B \;\middle|\; \begin{array}{l} \text{there exists some } a \in A \\ \text{such that } a \sim_R b \end{array} \right\}.$$

[1]Following **??**, **??**, we may compute the (characteristic functions associated to the) domain and range of a relation using the following colimit formulas:

$$\begin{aligned} \chi_{\mathrm{dom}(R)}(a) &\cong \operatorname*{colim}_{b \in B}\left(R_a^b\right) && (a \in A) \\ &\cong \bigvee_{b \in B} R_a^b, \\ \chi_{\mathrm{range}(R)}(b) &\cong \operatorname*{colim}_{a \in A}\left(R_a^b\right) && (b \in B) \\ &\cong \bigvee_{a \in A} R_a^b, \end{aligned}$$

where the join $\bigvee$ is taken in the poset $(\{\mathrm{true}, \mathrm{false}\}, \preceq)$ of Constructions With Sets, Definition 3.2.2.1.3.

[2]Viewing R as a function $R\colon A \to \mathcal{P}(B)$, we have

$$\begin{aligned} \mathrm{dom}(R) &\cong \operatorname*{colim}_{y \in Y}(R(y)) \\ &\cong \bigcup_{y \in Y} R(y), \\ \mathrm{range}(R) &\cong \operatorname*{colim}_{x \in X}(R(x)) \\ &\cong \bigcup_{x \in X} R(x), \end{aligned}$$

9.3.2 Binary Unions of Relations

Let A and B be sets and let R and S be relations from A to B.

Definition 9.3.2.1.1 ▶ Binary Unions of Relations

The **union of** R **and** S[1] is the relation $R \cup S$ from A to B defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[2]

$$R \cup S \stackrel{\text{def}}{=} \{(a, b) \in B \times A \mid \text{we have } a \sim_R b \text{ or } a \sim_S b\}.$$

- Viewing relations from A to B as functions $A \to \mathcal{P}(B)$, we define

$$[R \cup S](a) \stackrel{\text{def}}{=} R(a) \cup S(a)$$

  for each $a \in A$.

[1] *Further Terminology:* Also called the **binary union of** R **and** S, for emphasis.
[2] This is the same as the union of R and S as subsets of $A \times B$.

Proposition 9.3.2.1.2 ▶ Properties of Binary Unions of Relations

Let R, S, R_1, and R_2 be relations from A to B, and let S_1 and S_2 be relations from B to C.

1. *Interaction With Converses.* We have

$$(R \cup S)^\dagger = R^\dagger \cup S^\dagger.$$

2. *Interaction With Composition.* We have

$$(S_1 \diamond R_1) \cup (S_2 \diamond R_2) \stackrel{\text{poss.}}{\neq} (S_1 \cup S_2) \diamond (R_1 \cup R_2).$$

Proof 9.3.2.1.3 ▶ Proof of Proposition 9.3.2.1.2

Item 1: Interaction With Converses

Clear.

Item 2: Interaction With Composition

Unwinding the definitions, we see that:

- The condition for $(S_1 \diamond R_1) \cup (S_2 \diamond R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $b \sim_{S_1} c$;

    or
    - $a \sim_{R_2} b$ and $b \sim_{S_2} c$;
- The condition for $(S_1 \cup S_2) \diamond (R_1 \cup R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ or $a \sim_{R_2} b$;

    and
    - $b \sim_{S_1} c$ or $b \sim_{S_2} c$.

These two conditions may fail to agree (counterexample omitted), and thus the two resulting relations on $A \times C$ may differ.

9.3.3 Unions of Families of Relations

Let A and B be sets and let $\{R_i\}_{i \in I}$ be a family of relations from A to B.

Definition 9.3.3.1.1 ▶ The Union of a Family of Relations

The **union of the family** $\{R_i\}_{i \in I}$ is the relation $\bigcup_{i \in I} R_i$ from A to B defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[1]

$$\bigcup_{i \in I} R_i \stackrel{\text{def}}{=} \left\{ (a, b) \in (A \times B)^{\times I} \;\middle|\; \begin{array}{l} \text{there exists some } i \in I \\ \text{such that } a \sim_{R_i} b \end{array} \right\}.$$

- Viewing relations from A to B as functions $A \to \mathcal{P}(B)$, we define

$$\left[\bigcup_{i \in I} R_i \right](a) \stackrel{\text{def}}{=} \bigcup_{i \in I} R_i(a)$$

  for each $a \in A$.

[1] This is the same as the union of $\{R_i\}_{i \in I}$ as a collection of subsets of $A \times B$.

Proposition 9.3.3.1.2 ▶ Properties of Unions of Families of Relations

Let A and B be sets and let $\{R_i\}_{i\in I}$ be a family of relations from A to B.

1. *Interaction With Converses.* We have

$$\left(\bigcup_{i\in I} R_i\right)^{\dagger} = \bigcup_{i\in I} R_i^{\dagger}.$$

Proof 9.3.3.1.3 ▶ Proof of Proposition 9.3.3.1.2

Item 1: Interaction With Converses

Clear.

9.3.4 Binary Intersections of Relations

Let A and B be sets and let R and S be relations from A to B.

Definition 9.3.4.1.1 ▶ Binary Intersections of Relations

The **intersection of** R **and** S[1] is the relation $R \cap S$ from A to B defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[2]

$$R \cap S \stackrel{\text{def}}{=} \{(a, b) \in B \times A \mid \text{we have } a \sim_R b \text{ and } a \sim_S b\}.$$

- Viewing relations from A to B as functions $A \longrightarrow \mathcal{P}(B)$, we define

$$[R \cap S](a) \stackrel{\text{def}}{=} R(a) \cap S(a)$$

for each $a \in A$.

[1] *Further Terminology:* Also called the **binary intersection of** R **and** S, for emphasis.
[2] This is the same as the intersection of R and S as subsets of $A \times B$.

Proposition 9.3.4.1.2 ► Properties of Binary Intersections of Relations

Let R, S, R_1, and R_2 be relations from A to B, and let S_1 and S_2 be relations from B to C.

1. *Interaction With Converses.* We have

$$(R \cap S)^\dagger = R^\dagger \cap S^\dagger.$$

2. *Interaction With Composition.* We have

$$(S_1 \diamond R_1) \cap (S_2 \diamond R_2) = (S_1 \cap S_2) \diamond (R_1 \cap R_2).$$

Proof 9.3.4.1.3 ► Proof of Proposition 9.3.4.1.2

Item 1: Interaction With Converses

Clear.

Item 2: Interaction With Composition

Unwinding the definitions, we see that:

- The condition for $(S_1 \diamond R_1) \cap (S_2 \diamond R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $b \sim_{S_1} c$;

    and
    - $a \sim_{R_2} b$ and $b \sim_{S_2} c$;
- The condition for $(S_1 \cap S_2) \diamond (R_1 \cap R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $a \sim_{R_2} b$;

    and
    - $b \sim_{S_1} c$ and $b \sim_{S_2} c$.

These two conditions agree, and thus so do the two resulting relations on $A \times C$.

9.3.5 Intersections of Families of Relations

Let A and B be sets and let $\{R_i\}_{i\in I}$ be a family of relations from A to B.

Definition 9.3.5.1.1 ▶ The Intersection of a Family of Relations

The **intersection of the family** $\{R_i\}_{i\in I}$ is the relation $\bigcup_{i\in I} R_i$ defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[1]

$$\bigcup_{i\in I} R_i \stackrel{\text{def}}{=} \left\{ (a,b) \in (A\times B)^{\times I} \;\middle|\; \begin{array}{l} \text{for each } i \in I, \\ \text{we have } a \sim_{R_i} b \end{array} \right\}.$$

- Viewing relations from A to B as functions $A \to \mathcal{P}(B)$, we define

$$\left[\bigcap_{i\in I} R_i\right](a) \stackrel{\text{def}}{=} \bigcap_{i\in I} R_i(a)$$

  for each $a \in A$.

[1]This is the same as the intersection of $\{R_i\}_{i\in I}$ as a collection of subsets of $A \times B$.

Proposition 9.3.5.1.2 ▶ Properties of Intersections of Families of Relations

Let A and B be sets and let $\{R_i\}_{i\in I}$ be a family of relations from A to B.

1. *Interaction With Converses.* We have

$$\left(\bigcap_{i\in I} R_i\right)^{\dagger} = \bigcap_{i\in I} R_i^{\dagger}.$$

Proof 9.3.5.1.3 ▶ Proof of Proposition 9.3.5.1.2

Item 1: Interaction With Converses

Clear. ∎

9.3.6 Binary Products of Relations

Let A, B, X, and Y be sets, let $R\colon A \nrightarrow B$ be a relation from A to B, and let $S\colon X \nrightarrow Y$ be a relation from X to Y.

Definition 9.3.6.1.1 ► Binary Products of Relations

The **product of** R **and** S[1] is the relation $R \times S$ from $A \times X$ to $B \times Y$ defined as follows:

- Viewing relations from $A \times X$ to $B \times Y$ as subsets of $(A \times X) \times (B \times Y)$, we define $R \times S$ as the Cartesian product of R and S as subsets of $A \times X$ and $B \times Y$.[2]
- Viewing relations from $A \times X$ to $B \times Y$ as functions $A \times X \to \mathcal{P}(B \times Y)$, we define $R \times S$ as the composition

$$A \times X \xrightarrow{R \times S} \mathcal{P}(B) \times \mathcal{P}(Y) \overset{\mathcal{P}^{\otimes}_{B,Y}}{\hookrightarrow} \mathcal{P}(B \times Y)$$

  in Sets, i.e. by

$$[R \times S](a, x) \overset{\text{def}}{=} R(a) \times S(x)$$

  for each $(a, x) \in A \times X$.

[1] *Further Terminology:* Also called the **binary product of** R **and** S, for emphasis.

[2] That is, $R \times S$ is the relation given by declaring $(a, x) \sim_{R \times S} (b, y)$ iff $a \sim_R b$ and $x \sim_S y$.

Proposition 9.3.6.1.2 ► Properties of Binary Products of Relations

Let A, B, X, and Y be sets.

1. *Interaction With Converses.* Let

$$R\colon A \nrightarrow A,$$
$$S\colon X \nrightarrow X$$

   We have

$$(R \times S)^{\dagger} = R^{\dagger} \times S^{\dagger}.$$

2. *Interaction With Composition.* Let

$$\begin{aligned} R_1 &: A \nrightarrow B, \\ S_1 &: B \nrightarrow C, \\ R_2 &: X \nrightarrow Y, \\ S_2 &: Y \nrightarrow Z \end{aligned}$$

be relations. We have

$$(S_1 \diamond R_1) \times (S_2 \diamond R_2) = (S_1 \times S_2) \diamond (R_1 \times R_2).$$

Proof 9.3.6.1.3 ▶ Proof of Proposition 9.3.6.1.2

Item 1: Interaction With Converses

Unwinding the definitions, we see that:

- We have $(a, x) \sim_{(R \times S)^\dagger} (b, y)$ iff:
  - We have $(b, y) \sim_{R \times S} (a, x)$, i.e. iff:
    - We have $b \sim_R a$;
    - We have $y \sim_S x$;
- We have $(a, x) \sim_{R^\dagger \times S^\dagger} (b, y)$ iff:
  - We have $a \sim_{R^\dagger} b$ and $x \sim_{S^\dagger} y$, i.e. iff:
    - We have $b \sim_R a$;
    - We have $y \sim_S x$.

These two conditions agree, and thus the two resulting relations on $A \times X$ are equal.

Item 2: Interaction With Composition

Unwinding the definitions, we see that:

- We have $(a, x) \sim_{(S_1 \diamond R_1) \times (S_2 \diamond R_2)} (c, z)$ iff:

- We have $a \sim_{S_1 \diamond R_1} c$ and $x \sim_{S_2 \diamond R_2} z$, i.e. iff:
  - There exists some $b \in B$ such that $a \sim_{R_1} b$ and $b \sim_{S_1} c$;
  - There exists some $y \in Y$ such that $x \sim_{R_2} y$ and $y \sim_{S_2} z$;

- We have $(a, x) \sim_{(S_1 \times S_2) \diamond (R_1 \times R_2)} (c, z)$ iff:
  - There exists some $(b, y) \in B \times Y$ such that $(a, x) \sim_{R_1 \times R_2} (b, y)$ and $(b, y) \sim_{S_1 \times S_2} (c, z)$, i.e. such that:
    - We have $a \sim_{R_1} b$ and $x \sim_{R_2} y$;
    - We have $b \sim_{S_1} c$ and $y \sim_{S_2} z$.

These two conditions agree, and thus the two resulting relations from $A \times X$ to $C \times Z$ are equal.

9.3.7 Products of Families of Relations

Let $\{A_i\}_{i \in I}$ and $\{B_i\}_{i \in I}$ be families of sets, and let $\{R_i \colon A_i \nrightarrow B_i\}_{i \in I}$ be a family of relations.

Definition 9.3.7.1.1 ► The Product of a Family of Relations

The **product of the family** $\{R_i\}_{i \in I}$ is the relation $\prod_{i \in I} R_i$ from $\prod_{i \in I} A_i$ to $\prod_{i \in I} B_i$ defined as follows:

- Viewing relations as subsets, we define $\prod_{i \in I} R_i$ as its product as a family of sets, i.e. we have

$$\prod_{i \in I} R_i \stackrel{\text{def}}{=} \left\{ (a_i, b_i)_{i \in I} \in \prod_{i \in I} (A_i \times B_i) \;\middle|\; \begin{array}{l} \text{for each } i \in I, \\ \text{we have } a_i \sim_{R_i} b_i \end{array} \right\}.$$

- Viewing relations as functions to powersets, we define

$$\left[\prod_{i \in I} R_i \right] ((a_i)_{i \in I}) \stackrel{\text{def}}{=} \prod_{i \in I} R_i(a_i)$$

  for each $(a_i)_{i \in I} \in \prod_{i \in I} R_i$.

9.3.8 The Collage of a Relation

Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation from A to B.

Definition 9.3.8.1.1 ► The Collage of a Relation

The **collage of** R[1] is the poset $\mathbf{Coll}(R) \overset{\text{def}}{=} \left(\mathrm{Coll}(R), \preceq_{\mathbf{Coll}(R)}\right)$ consisting of:

- *The Underlying Set.* The set $\mathrm{Coll}(R)$ defined by

$$\mathrm{Coll}(R) \overset{\text{def}}{=} A \coprod B.$$

- *The Partial Order.* The partial order

$$\preceq_{\mathbf{Coll}(R)}\colon \mathrm{Coll}(R) \times \mathrm{Coll}(R) \longrightarrow \{\mathsf{true}, \mathsf{false}\}$$

on $\mathrm{Coll}(R)$ defined by

$$\preceq(a, b) \overset{\text{def}}{=} \begin{cases} \mathsf{true} & \text{if } a = b \text{ or } a \sim_R b, \\ \mathsf{false} & \text{otherwise.} \end{cases}$$

[1] *Further Terminology:* Also called the **cograph of** R.

Notation 9.3.8.1.2 ► Notation: $\mathsf{Pos}_{/\Delta^1}(A, B)$

We write $\mathsf{Pos}_{/\Delta^1}(A, B)$ for the category defined as the pullback

$$\mathsf{Pos}_{/\Delta^1}(A, B) \overset{\text{def}}{=} \mathsf{pt} \underset{[A],\mathsf{Pos},\mathsf{ev}_0}{\times} \mathsf{Pos}_{/\Delta^1} \underset{\mathsf{ev}_1,\mathsf{Pos},[B]}{\times} \mathsf{pt},$$

as in the diagram

$$\begin{tikzcd}
& & \mathbf{Pos}_{/\Delta^1}(A,B) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\vee"] & & \\
& \mathbf{Pos}_{/\Delta^1} \underset{\mathbf{Pos}}{\times} \mathrm{pt} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\vee"] & & \mathrm{pt} \underset{\mathbf{Pos}}{\times} \mathbf{Pos}_{/\Delta^1} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\vee"] & \\
\mathrm{pt} \arrow[dr, "{[A]}"'] & & \mathbf{Pos}_{/\Delta^1} \arrow[dl, "\mathrm{ev}_{[0]}"] \arrow[dr, "\mathrm{ev}_{[1]}"'] & & \mathrm{pt}. \arrow[dl, "{[B]}"] \\
& \mathbf{Pos} & & \mathbf{Pos} &
\end{tikzcd}$$

Remark 9.3.8.1.3 ▶ Unwinding Notation 9.3.8.1.2

In detail, $\mathrm{Pos}_{/\Delta^1}(A,B)$ is the category where:

- *Objects.* An object of $\mathrm{Pos}_{/\Delta^1}(A,B)$ is a pair (X,ϕ_X) consisting of
  - A poset X;
  - A morphism $\phi_X\colon X \to \Delta^1$;

  such that we have

$$\begin{aligned}\phi_X^{-1}(0) &= A,\\ \phi_X^{-1}(1) &= B.\end{aligned}$$

- *Morphisms.* A morphism of $\mathrm{Pos}_{/\Delta^1}(A,B)$ from (X,ϕ_X) to (Y,ϕ_Y) is a morphism of posets $f\colon X \to Y$ making the diagram

$$\begin{tikzcd}
X \arrow[rr, "f"] \arrow[dr, "\phi_X"'] & & Y \arrow[dl, "\phi_Y"] \\
& \Delta^1 &
\end{tikzcd}$$

  commute.

Proposition 9.3.8.1.4 ► Properties of Collages of Relations

Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation from A to B.

1. *Functoriality.* The assignment $R \mapsto \mathbf{Coll}(R)$ defines a functor

$$\mathbf{Coll}\colon \mathbf{Rel}(A,B) \longrightarrow \mathsf{Pos}_{/\Delta^1}(A,B),$$

where

- *Action on Objects.* For each $R \in \mathrm{Obj}(\mathbf{Rel}(A,B))$, we have

$$[\mathbf{Coll}](R) \overset{\text{def}}{=} (\mathbf{Coll}(R), \phi_R)$$

for each $R \in \mathbf{Rel}(A,B)$, where

  - The poset $\mathbf{Coll}(R)$ is the collage of R of Definition 9.3.8.1.1.
  - The morphism $\phi_R\colon \mathbf{Coll}(R) \longrightarrow \Delta^1$ is given by

$$\phi_R(x) \overset{\text{def}}{=} \begin{cases} 0 & \text{if } x \in A, \\ 1 & \text{if } x \in B \end{cases}$$

for each $x \in \mathbf{Coll}(R)$.

- *Action on Morphisms.* For each $R, S \in \mathrm{Obj}(\mathbf{Rel}(A,B))$, the action on Hom-sets

$$\mathbf{Coll}_{R,S}\colon \mathrm{Hom}_{\mathbf{Rel}(A,B)}(R,S) \longrightarrow \mathsf{Pos}(\mathbf{Coll}(R), \mathbf{Coll}(S))$$

of $\mathbf{Coll}$ at (R,S) is given by sending an inclusion

$$\iota\colon R \subset S$$

to the morphism

$$\mathbf{Coll}(\iota)\colon \mathbf{Coll}(R) \longrightarrow \mathbf{Coll}(S)$$

of posets over Δ^1 defined by

$$[\mathbf{Coll}(\iota)](x) \overset{\text{def}}{=} x$$

for each $x \in \mathbf{Coll}(R)$.[1]

2. *Equivalence.* The functor of Item 1 is an equivalence of categories.

[1] Note that this is indeed a morphism of posets: if $x \preceq_{\mathbf{Coll}(R)} y$, then $x = y$ or $x \sim_R y$, so we have either $x = y$ or $x \sim_S y$ (as $R \subset S$), and thus $x \preceq_{\mathbf{Coll}(S)} y$.

Proof 9.3.8.1.5 ▶ Proof of Proposition 9.3.8.1.4

Item 1: Functoriality

Clear.

Item 2: Equivalence

Omitted.

Appendices

A Other Chapters

Preliminaries

1. Introduction
2. A Guide to the Literature

Sets

3. Sets
4. Constructions With Sets
5. Monoidal Structures on the Category of Sets
6. Pointed Sets
7. Tensor Products of Pointed Sets

Relations

8. Relations
9. Constructions With Relations
10. Conditions on Relations

Categories

11. Categories

Monoidal Categories

12. Constructions With Monoidal Categories

Bicategories

13. Types of Morphisms in Bicategories

Extra Part

14. Notes

References

[MO 460656] Emily. *Existence and characterisations of left Kan extensions and liftings in the bicategory of relations I*. MathOverflow. url: https://mathoverflow.net/q/460656 (cit. on pp. 3, 4).

[MO 461592] Emily. *Existence and characterisations of left Kan extensions and liftings in the bicategory of relations II*. MathOverflow. url: https://mathoverflow.net/q/461592 (cit. on pp. 3, 4, 6).